



Experiment 8

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Branch: CSE

Semester: 6th

Subject Name: Full Stack Development – II

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Section/Group: KRG 3-B

Date of Performance: 31/03/2026

Subject Code: 23CSH-309

1. Aim: To design and implement a Spring Boot application that integrates third-party APIs including **Stripe Payment Gateway** for handling online transactions and **Gmail SMTP Service** for sending email notifications after successful payments.

2. Objective:

- To understand the concept of third-party API integration
- To integrate Stripe Payment Gateway for secure online payments
- To configure and use Gmail SMTP for sending emails
- To implement REST APIs for payment processing
- To handle payment success and failure responses
- To send automated email notifications after successful payment
- To follow Layered Architecture (Controller – Service – Repository)
- To enhance real-world application functionality using external services

3. Implementation / Code:

▪ Tools & Technologies Used:-

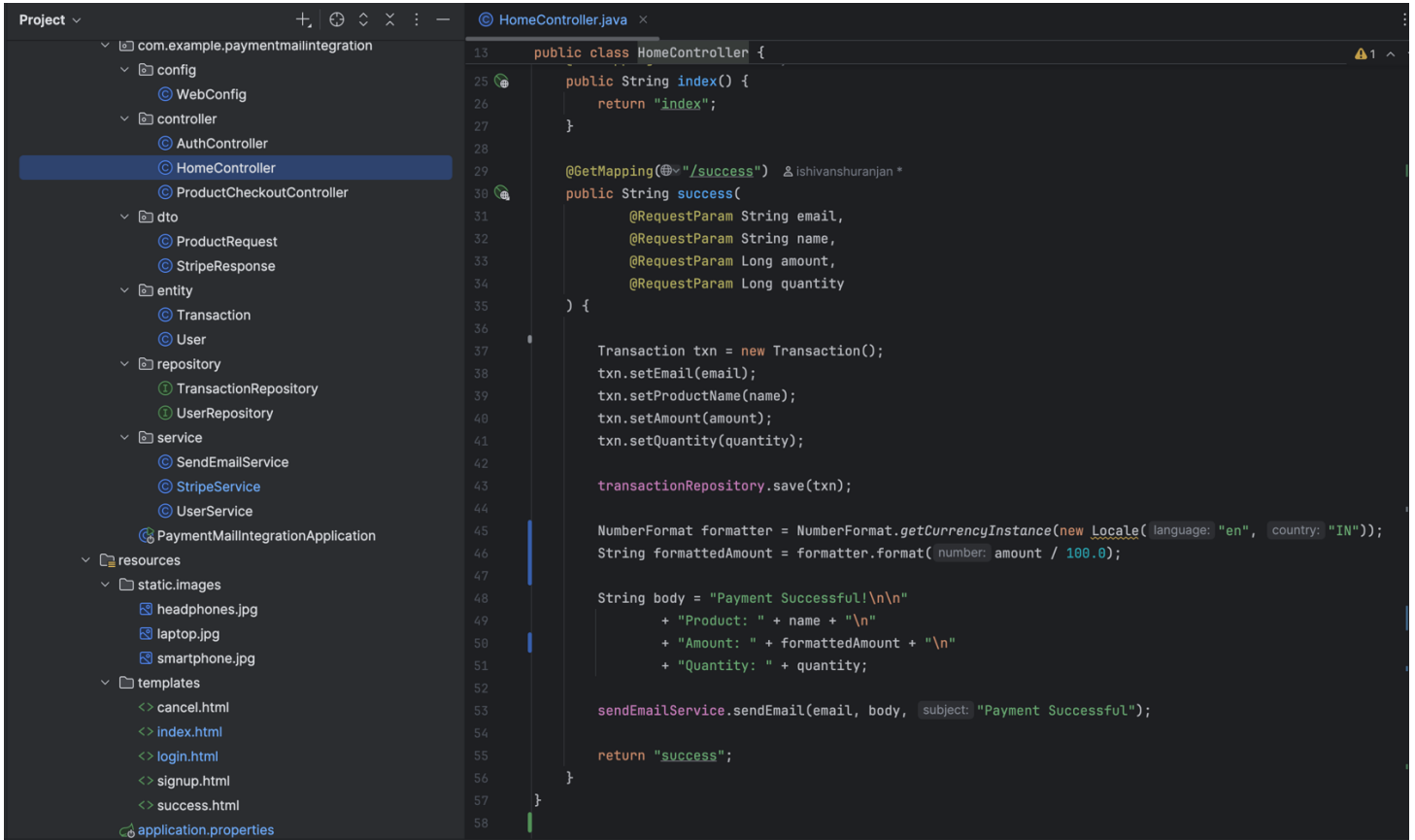
- Java
- Spring Boot
- Spring Web
- Spring Data JPA
- Maven
- IntelliJ IDEA
- Stripe API
- Gmail SMTP (Java Mail Sender)
- Postman / Browser
- PostgreSQL Database

▪ Implementation Description:-

- A Spring Boot project named Payment Mail Integration is created.
- The project integrates Stripe API for handling online payments securely.
- Users enter card details on Stripe Checkout page and complete payment.

- After successful payment, a success page is displayed.
- A confirmation email is automatically sent to the user using Gmail SMTP.

■ Project Structure:-



The screenshot shows an IDE with a project structure on the left and the `HomeController.java` file open in the editor. The project structure includes:

- com.example.paymentmailintegration
 - config
 - WebConfig
 - controller
 - AuthController
 - HomeController
 - ProductCheckoutController
 - dto
 - ProductRequest
 - StripeResponse
 - entity
 - Transaction
 - User
 - repository
 - TransactionRepository
 - UserRepository
 - service
 - SendEmailService
 - StripeService
 - UserService
 - PaymentMailIntegrationApplication
 - resources
 - static.images
 - headphones.jpg
 - laptop.jpg
 - smartphone.jpg
 - templates
 - cancel.html
 - index.html
 - login.html
 - signup.html
 - success.html
 - application.properties

The `HomeController.java` file contains the following code:

```
13 public class HomeController {
25     public String index() {
26         return "index";
27     }
28
29     @GetMapping("/success") & ishivanshuranjan *
30     public String success(
31         @RequestParam String email,
32         @RequestParam String name,
33         @RequestParam Long amount,
34         @RequestParam Long quantity
35     ) {
36
37         Transaction txn = new Transaction();
38         txn.setEmail(email);
39         txn.setProductName(name);
40         txn.setAmount(amount);
41         txn.setQuantity(quantity);
42
43         transactionRepository.save(txn);
44
45         NumberFormat formatter = NumberFormat.getCurrencyInstance(new Locale("en", "IN"));
46         String formattedAmount = formatter.format(number: amount / 100.0);
47
48         String body = "Payment Successful!\n\n"
49             + "Product: " + name + "\n"
50             + "Amount: " + formattedAmount + "\n"
51             + "Quantity: " + quantity;
52
53         sendEmailService.sendEmail(email, body, subject: "Payment Successful");
54
55         return "success";
56     }
57 }
58
```

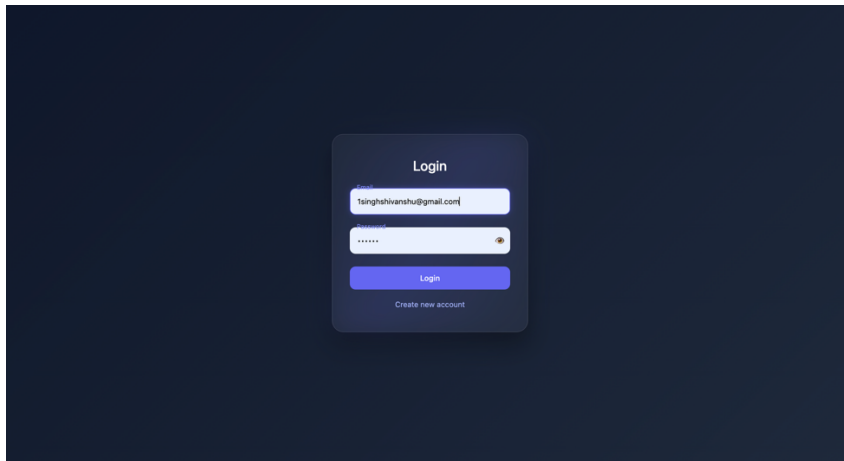
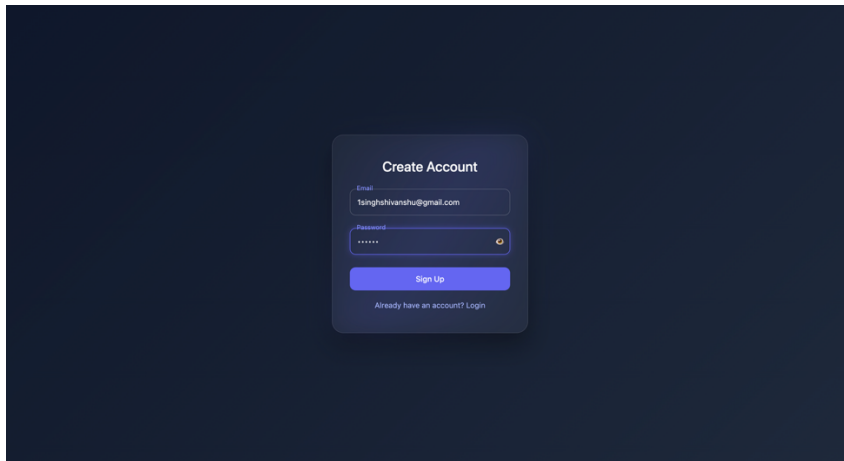
4. Output:

- Spring Boot application started successfully on **localhost:8080**
- Stripe payment page loaded successfully
- Payment completed successfully using test card
- Success page displayed after payment completion
- Email notification sent successfully to registered email
- Email contains product details, amount, and quantity
- Payment flow tested successfully using Stripe test environment
- Application integrated successfully with third-party APIs



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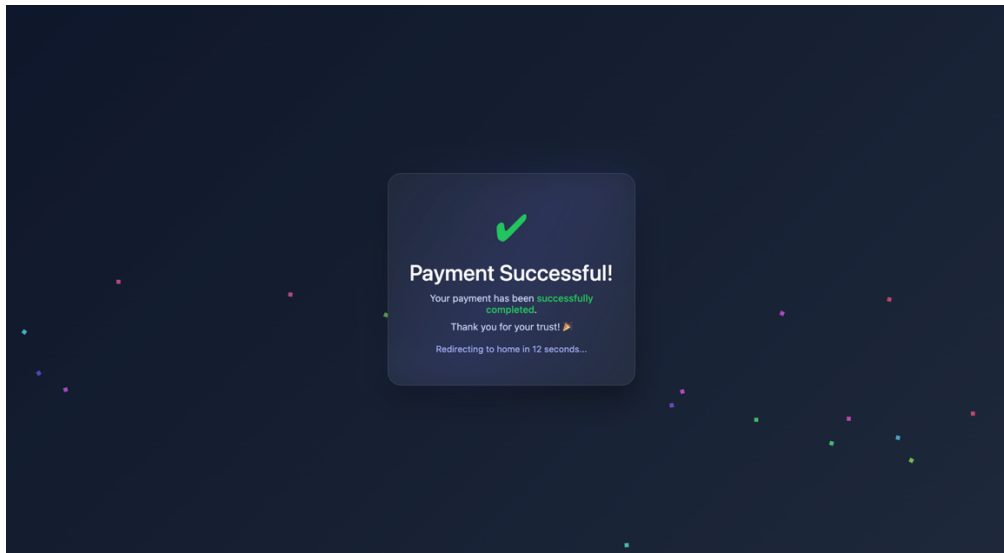
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to me ▼

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Product: Headphones

Amount: ₹50,000.00

Quantity: 2

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➡ Forward



5. Learning Outcomes (What I Have Learnt):

- Understanding of third-party API integration
- Working with Stripe Payment Gateway
- Secure handling of online transactions
- Sending emails using Gmail SMTP
- Implementing real-world payment systems
- Integration of frontend and backend systems
- Usage of REST APIs in real applications
- Handling success and failure scenarios
- Improving application functionality using external services